

OLAP IN THE DATA WAREHOUSE

CHAPTER OBJECTIVES

- Perceive the unqualified demand for online analytical processing (OLAP) and understand what drives this demand
- Review the major features and functions of OLAP in detail
- Grasp the intricacies of dimensional analysis and learn the meanings of hypercubes, drill-down and roll-up, and slice-and-dice
- Examine the different OLAP models and determine which model is suitable for your environment
- Consider OLAP implementation by studying the steps and the tools

In the earlier chapters we mentioned online analytical processing (OLAP) in passing. You had a glimpse of OLAP when we discussed the information delivery methods. You have some idea of what OLAP is and how it is used for complex analysis. As the name implies, OLAP has to do with the processing of data as it is manipulated for analysis. The data warehouse provides the best opportunity for analysis and OLAP is the vehicle for carrying out involved analysis. The data warehouse environment is also best for data access when analysis is carried out.

We now have the chance to explore OLAP in sufficient depth. In today's data warehousing environment, with such tremendous progress in analysis tools from various vendors, you cannot have a data warehouse without OLAP. It is unthinkable. Therefore, throughout this chapter, look out for the important topics in OLAP.

First, you have to perceive what OLAP is and why it is absolutely essential. This will help you to better understand the features and functions of OLAP. We will discuss the major features and functions so that your grasp of OLAP may be firmed up. There are two

major models for OLAP. You should know which model is most suitable for your computing and user environments. We will highlight the significance of each model, learn how to implement OLAP in your data warehouse environment, investigate OLAP tools, and find out how to evaluate and procure them for your users. Finally, we will discuss the implementation steps for OLAP.

DEMAND FOR ONLINE ANALYTICAL PROCESSING

Recall our discussions in Chapter 2 of the top-down and bottom-up approaches for building a data warehouse. In the top-down approach, you build the overall corporate-wide data repository using the entity-relationship (E-R) data modeling technique. This enterprise-wide data warehouse feeds the departmental data marts that are designed using the dimensional modeling technique. In the bottom-up approach, you build several data marts using the dimensional modeling technique and the collection of these data marts forms the data warehouse environment for your company. Each of these two approaches has its advantages and shortcomings.

You also learned about a practical approach to building a conglomeration of supermarts with conformed and standardized data content. While adopting this approach, first you plan and define the requirements at the corporate level, build the infrastructure for the complete warehouse, and then implement one supermart at a time in a priority sequence. The supermarts are designed using the dimensional modeling technique.

As we have seen, a data warehouse is meant for performing substantial analysis using the available data. The analysis leads to strategic decisions that are the major reasons for building data warehouses in the first place. For performing meaningful analysis, data must be cast in a way suitable for analysis of the values of key indicators over time along business dimensions. Data structures designed using the dimensional modeling technique support such analysis.

In the three approaches referred to above, the data marts rest on the dimensional model. Therefore, these data marts must be able to support dimensional analysis. In practice, these data marts seem to be adequate for basic analysis. However, in today's business conditions, we find that users need to go beyond such basic analysis. They must have the capability to perform far more complex analysis in less time. Let us examine how the traditional methods of analysis provided in a data warehouse are not sufficient and perceive what exactly is demanded by the users to stay competitive and to expand.

Need for Multidimensional Analysis

Let us quickly review the business model of a large retail operation. If you just look at daily sales, you soon realize that the sales are interrelated with many business dimensions. The daily sales are meaningful only when they are related to the dates of the sales, the products, the distribution channels, the stores, the sales territories, the promotions, and a few more dimensions. Multidimensional views are inherently representative of any business model. Very few models are limited to three dimensions or less. For planning and making strategic decisions, managers and executives probe into business data through scenarios. For example, they compare actual sales against targets and against sales in prior periods. They examine the breakdown of sales by product, by store, by sales territory, by promotion, and so on.

Decision makers are no longer satisfied with one-dimensional queries such as "How many units of product A did we sell in the store in Edison, New Jersey?" Consider the

following more useful query: “How much revenue did the new product X generate during the last three months, broken down by individual months, in the south central territory, by individual stores, broken down by promotions, compared to estimates, and compared to the previous version of the product?” The analysis does not stop with this single multi-dimensional query. The user continues to ask for further comparisons to similar products, comparisons among territories, and views of the results by rotating the presentation between columns and rows.

For effective analysis, your users must have easy methods of performing complex analysis along several business dimensions. They need an environment that presents a multi-dimensional view of data, providing the foundation for analytical processing through easy and flexible access to information. Decision makers must be able to analyze data along any number of dimensions, at any level of aggregation, with the capability of viewing results in a variety of ways. They must have the ability to drill down and roll up along the hierarchies of every dimension. Without a solid system for true multidimensional analysis, your data warehouse is incomplete.

In any analytical system, time is a critical dimension. Hardly any query is executed without having time as one of the dimensions along which analysis is performed. Further, time is a unique dimension because of its sequential nature—November always comes after October. Users monitor performance over time, as for example, performance this month compared to last month, or performance this month compared with performance the same month last year.

Another point about the uniqueness of the time dimension is the way in which the hierarchies of the dimension work. A user may look for sales in March and may also look for sales for the first four months of the year. In the second query for sales for the first four months, the implied hierarchy at the next higher level is an aggregation taking into account the sequential nature of time. No user looks for sales of the first four stores or the last three stores. There is no implied sequence in the store dimension. True analytical systems must recognize the sequential nature of time.

Fast Access and Powerful Calculations

Whether a user’s request is for monthly sales of all products along all geographical regions or for year-to-date sales in a region for a single product, the query and analysis system must have consistent response times. Users must not be penalized for the complexity of their analysis. Both the size of the effort to formulate a query or the amount of time to receive the result sets must be consistent irrespective of the query types.

Let us take an example to understand how speed of the analysis process matters to users. Imagine a business analyst looking for reasons why profitability dipped sharply in recent months in the entire enterprise. The analyst starts this analysis by querying for the overall sales for the last five months for the entire company, broken down by individual months. The analyst notices that although the sales do not show a drop, there is a sharp reduction in profitability for the last three months. The analysis proceeds further when the analyst wants to find out which countries show reductions. The analyst requests a breakdown of sales by major worldwide regions and notes that the European region is responsible for the reduction in profitability. Now the analyst senses that clues are becoming more pronounced and looks for a breakdown of the European sales by individual countries. The analyst finds that the profitability has increased for a few countries, decreased sharply for some other countries, and been stable for the rest. At this point, the analyst introduces another

dimension into the analysis. Now the analyst wants the breakdown of profitability for the European countries by country, month, and product. This step brings the analyst closer to the reason for the decline in the profitability. The analyst observes that the countries in the European Union (EU) show very sharp declines in profitability for the last two months. Further queries reveal that manufacturing and other direct costs remain at the usual levels but the indirect costs have shot up. The analyst is now able to determine that the decline is due to the additional tax levies on some products in the EU. The analyst has also determined the exact effect of the levies so far. Strategic decisions follow on how to deal with the decline in profitability.

Now look at Figure 15-1 showing the steps through the single analysis session. How many steps are there? Many steps, but one single analysis session with a necessary continuous train of thought. Each step in this train of thought constitutes a query. The analyst formulates each query, executes it, waits for the result set to appear on the screen, and studies the result set. Each query is interactive because the result set from one query forms the basis for the next query. In this manner of querying, the user cannot maintain the train of thought unless the momentum is preserved. Fast access is absolutely essential for an effective analytical processing environment.

Did you notice that none of the queries in the above analysis session included any serious calculations? This is not typical. In a real-world analysis session, many of the queries require calculations, sometimes complex calculations. What is the implication here? An effective analytical processing environment must not only be fast and flexible, but it must also support complex and powerful calculations.

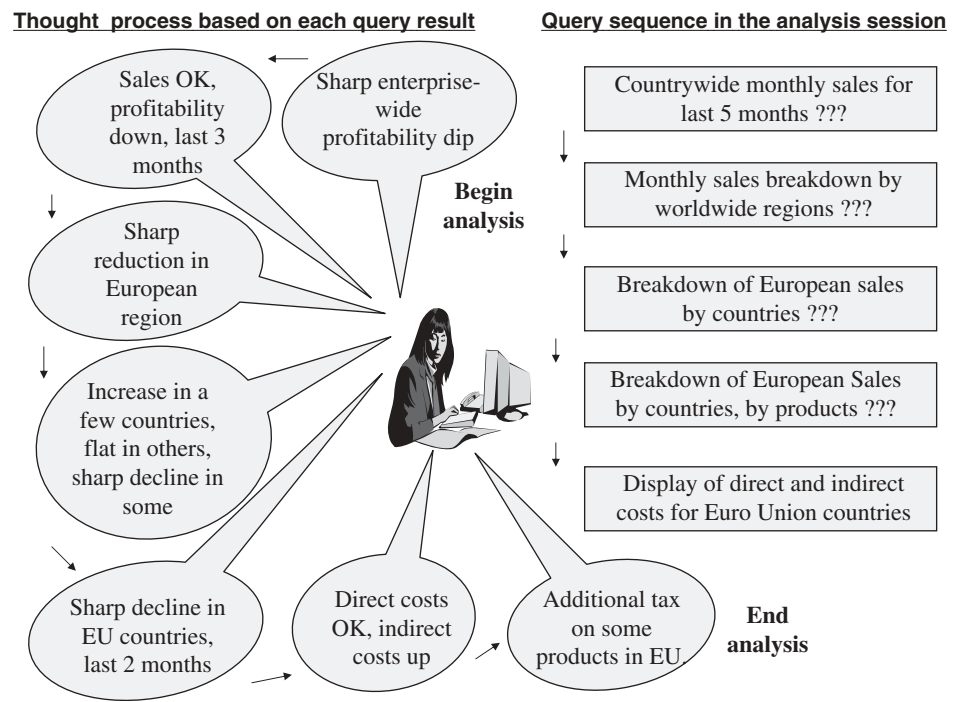


Figure 15-1 Query steps in an analysis session.

What follows is a list of typical calculations that get included in the query requests:

- Roll ups to provide summaries and aggregations along the hierarchies of the dimensions
- Drill downs from the top level to the lowest along the hierarchies of the dimensions, in combinations among the dimensions
- Simple calculations, such as computation of margins (sales minus costs)
- Share calculations to compute the percentage of parts to the whole
- Algebraic equations involving key performance indicators
- Moving averages and growth percentages
- Trend analysis using statistical methods

Limitations of Other Analysis Methods

You now have a fairly good grip on the types of requirements of users to execute queries and perform analysis. First and foremost, the information delivery system must be able to present multidimensional views of the data. Then the information delivery system must enable the users to use the data by analyzing it along multiple dimensions and their hierarchies in a myriad of ways. And this facility must be fast. It must be possible for users to perform complex calculations.

Let us understand why the traditional tools and methods are not up to the task when it comes to complex analysis and calculations. What information methods are we familiar with? Of course, the earliest method was the medium of reports. Then came spreadsheets with all their functionality and features. SQL has been the accepted interface for retrieving and manipulating data from relational databases. These methods are used in OLTP systems and in data warehouse environments. Now, when we discuss multidimensional analysis and complex calculations, how suitable are these traditional methods?

First, let us compare the characteristics of the OLTP and data warehouse environments. When we mention the data warehouse environment here, we are not referring to heavy multidimensional analysis and complex calculations. We are only referring to the environment with simple queries and routine reports. Figure 15-2 shows the characteristics of the OLTP and the basic data warehouse environments as they relate to information delivery needs.

Now consider information retrieval and manipulation in these two environments. What are the standard methods of information delivery? These are reports, spreadsheets, and online displays. What is the standard data access interface? SQL. Let us review these and determine if they are adequate for multidimensional analysis and complex calculations.

Report writers provide two key functions: the ability to point and click for generating and issuing SQL calls, and the capability to format the output reports. However, report writers do not support multidimensionality. With basic report writers, you cannot drill down to lower levels in the dimensions. That will have to come from additional reports. You cannot rotate the results by switching rows and columns. The report writers do not provide aggregate navigation. Once the report is formatted and run, you cannot alter the presentation of the result data sets.

If report writers are not the tools or methods we are looking for, how about spreadsheets for calculations and the other features needed for analysis? Spreadsheets, when they first appeared, were positioned as analysis tools. You can perform “what if” analysis with

CHARACTERISTICS	OLTP SYSTEMS	DATA WAREHOUSE
Analytical capabilities	Very low	Moderate
Data for a single session	Very limited	Small to medium size
Size of result set	Small	Large
Response time	Very fast	Fast to moderate
Data granularity	Detail	Detail and summary
Data currency	Current	Current and historical
Access method	Predefined	Predefined and ad hoc
Basic motivation	Collect and input data	Provide information
Data model	Design for data updates	Design for queries
Optimization of database	For transactions	For analysis
Update frequency	Very frequent	Generally read-only
Scope of user interaction	Single transactions	Throughout data content

Figure 15-2 OLTP and data warehouse environments.

spreadsheets. When you modify the values in some cells, the values in other related cells automatically change. What about aggregations and calculations?

Spreadsheets with their add-in tools can perform some forms of aggregations and also do a variety of calculations. Third-party tools have also enhanced spreadsheet products to present data in three-dimensional formats. You can view rows, columns, and pages on spreadsheets. For example, the rows can represent products, the columns represent stores, and the pages represent the time dimension in months. Modern spreadsheet tools offer pivot tables or n-way cross-tabs.

Even with enhanced functionality using add-ins, spreadsheets are still very cumbersome to use. Take an analysis involving the four dimensions of store, product, promotion, and time. Let us say each dimension contains an average of five hierarchical levels. Now try to build an analysis to retrieve data and present it as spreadsheets showing all the aggregation levels and multidimensional views, and also using even simple calculations. You can very well imagine how much effort it would take for this exercise. Now what if your user wants to change the navigation and do different roll ups and drill downs. The limitations of spreadsheets for multidimensional analysis and complex calculations are quite evident.

Let us now turn our attention to SQL (Structured Query Language). Although it might have been the original goal of SQL to be the end-user query language, now everyone agrees that the language is too abstruse even for sophisticated users. Third-party products attempt to extend the capabilities of SQL and hide the syntax from the users. Users can formulate their queries through GUI point-and-click methods or by using natural language syntax. Nevertheless, SQL vocabulary is ill-suited for analyzing data and exploring relationships. Even basic comparisons prove to be difficult in SQL.

Meaningful analysis such as market exploration and financial forecasting typically involve retrieval of large quantities of data, performing calculations, and summarizing the data on the fly. Perhaps, even the detailed analysis may be achieved by using SQL for

retrieval and spreadsheets for presenting the results. But here is the catch: in a real-world analysis session, many queries follow one after the other. Each query may translate into a number of intricate SQL statements, with each of the statements likely to invoke full table scans, multiple joins, aggregations, groupings, and sorting. Analysis of the type we are discussing requires complex calculations and handling time series data. SQL is notably weak in these areas. Even if you can imagine an analyst accurately formulating such complex SQL statements, the overhead on the systems would still be enormous and seriously impact response times.

OLAP is the Answer

Users certainly need the ability to perform multidimensional analysis with complex calculations, but we find that the traditional tools of report writers, query products, spreadsheets, and language interfaces are distressfully inadequate. What is the answer? Clearly, the tools being used in the OLTP and basic data warehouse environments do not match up to the task. We need a different set of tools and products that are specifically meant for serious analysis. We need OLAP in the data warehouse.

In this chapter, we will thoroughly examine the various aspects of OLAP. We will come up with formal definitions and detailed characteristics. We will highlight all the features and functions. We will explore the different OLAP models. But now that you have an initial appreciation for OLAP, let us list the basic virtues of OLAP to justify our proposition.

- Enables analysts, executives, and managers to gain useful insights from the presentation of data.
- Can reorganize metrics along several dimensions and allow data to be viewed from different perspectives.
- Supports multidimensional analysis.
- Is able to drill down or roll up within each dimension.
- Is capable of applying mathematical formulas and calculations to measures.
- Provides fast response, facilitating speed-of-thought analysis.
- Complements the use of other information delivery techniques such as data mining.
- Improves the comprehension of result sets through visual presentations using graphs and charts.
- Can be implemented on the Web.
- Designed for highly interactive analysis.

Even at this stage, you will further appreciate the nature and strength of OLAP by studying a typical OLAP session (see Fig. 15-3). The analyst starts with a query requesting a high-level summary by product line. Next, the user moves to drilling down for details by year. In the following step, the analyst pivots the data to view totals by year rather than totals by product line. Even in such a simple example, you observe the power and features of OLAP.

OLAP Definitions and Rules

Where did the term OLAP originate? We know that multidimensionality is at the core of OLAP systems. We have also mentioned some other basic features of OLAP. Is it a vague collection of complex factors for serious analysis? Is there a formal definition and a set of fundamental guidelines identifying OLAP systems?

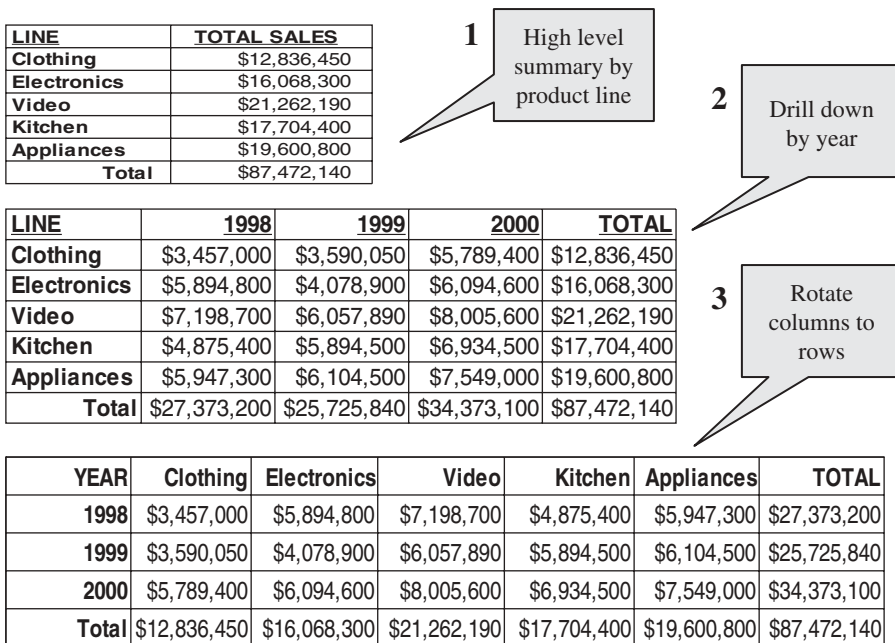


Figure 15-3 Simple OLAP session.

The term OLAP or online analytical processing was introduced in a paper entitled “Providing On-Line Analytical Processing to User Analysts,” by Dr. E. F. Codd, the acknowledged father of the relational database model. The paper, published in 1993, defined 12 rules or guidelines for an OLAP system. Later, in 1995, six additional rules were included. We will discuss these rules. Before that, let us look for a short and precise definition for OLAP. Such a succinct definition comes from the OLAP council (www.olapcouncil.org), which provides membership, sponsors research, and promotes the use of OLAP. Here is the definition:

On-Line Analytical Processing (OLAP) is a category of software technology that enables analysts, managers and executives to gain insight into data through fast, consistent, interactive access in a wide variety of possible views of information that has been transformed from raw data to reflect the real dimensionality of the enterprise as understood by the user.

The definition from the OLAP council contains all the key ingredients. Speed, consistency, interactive access, and multiple dimensional views—all of these are principal elements. As one trade magazine described it as early as in 1995, OLAP is a fancy term for multidimensional analysis.

The guidelines proposed by Dr. Codd form the yardstick for measuring any sets of OLAP tools and products. A true OLAP system must conform to these guidelines. When your project team is looking for OLAP tools, it can prioritize these guidelines and select tools that meet the set of criteria at the top of your priority list. First, let us consider the initial 12 guidelines for an OLAP system:

Multidimensional Conceptual View. Provide a multidimensional data model that is intuitively analytical and easy to use. Business users’ view of an enterprise is

multidimensional in nature. Therefore, a multidimensional data model conforms to how the users perceive business problems.

Transparency. Make the technology, underlying data repository, computing architecture, and the diverse nature of source data totally transparent to users. Such transparency, supporting a true open system approach, helps to enhance the efficiency and productivity of the users through front-end tools that are familiar to them.

Accessibility. Provide access only to the data that is actually needed to perform the specific analysis, presenting a single, coherent, and consistent view to the users. The OLAP system must map its own logical schema to the heterogeneous physical data stores and perform any necessary transformations.

Consistent Reporting Performance. Ensure that the users do not experience any significant degradation in reporting performance as the number of dimensions or the size of the database increases. Users must perceive consistent run time, response time, or machine utilization every time a given query is run.

Client/Server Architecture. Conform the system to the principles of client/server architecture for optimum performance, flexibility, adaptability, and interoperability. Make the server component sufficiently intelligent to enable various clients to be attached with a minimum of effort and integration programming.

Generic Dimensionality. Ensure that every data dimension is equivalent in both structure and operational capabilities. Have one logical structure for all dimensions. The basic data structure or the access techniques must not be biased toward any single data dimension.

Dynamic Sparse Matrix Handling. Adapt the physical schema to the specific analytical model being created and loaded that optimizes sparse matrix handling. When encountering a sparse matrix, the system must be able to dynamically deduce the distribution of the data and adjust the storage and access to achieve and maintain consistent level of performance.

Multiuser Support. Provide support for end users to work concurrently with either the same analytical model or to create different models from the same data. In short, provide concurrent data access, data integrity, and access security.

Unrestricted Cross-Dimensional Operations. Provide ability for the system to recognize dimensional hierarchies and automatically perform roll-up and drill-down operations within a dimension or across dimensions. Have the interface language allow calculations and data manipulations across any number of data dimensions, without restricting any relations between data cells, regardless of the number of common data attributes each cell contains.

Intuitive Data Manipulation. Enable consolidation path reorientation (pivoting), drill down and roll up, and other manipulations to be accomplished intuitively and directly via point-and-click and drag-and-drop actions on the cells of the analytical model. Avoid the use of a menu or multiple trips to a user interface.

Flexible Reporting. Provide capabilities to the business user to arrange columns, rows, and cells in a manner that facilitates easy manipulation, analysis, and synthesis of information. Every dimension, including any subsets, must be able to be displayed with equal ease.

Unlimited Dimensions and Aggregation Levels. Accommodate at least 15, preferably 20, data dimensions within a common analytical model. Each of these generic dimensions

must allow a practically unlimited number of user-defined aggregation levels within any given consolidation path.

In addition to these 12 basic guidelines, also take into account the following requirements, not all distinctly specified by Dr. Codd.

Drill-Through to Detail Level. Allow a smooth transition from the multidimensional, pre-aggregated database to the detail record level of the source data warehouse repository.

OLAP Analysis Models. Support Dr. Codd's four analysis models: exegetical (or descriptive), categorical (or explanatory), contemplative, and formulaic.

Treatment of Nonnormalized Data. Prohibit calculations made within an OLAP system from affecting the external data serving as the source.

Storing OLAP Results. Do not deploy write-capable OLAP tools on top of transactional systems.

Missing Values. Ignore missing values, irrespective of their source.

Incremental Database Refresh. Provide for incremental refreshes of the extracted and aggregated OLAP data.

SQL Interface. Seamlessly integrate the OLAP system into the existing enterprise environment.

OLAP Characteristics

Let us summarize in simple terms what we have covered so far. We explored why business users absolutely need online analytical processing. We examined why the other methods of information delivery do not satisfy the requirements for multidimensional analysis with powerful calculations and fast access. We discussed how OLAP is the answer to satisfy these requirements. We reviewed the definitions and authoritative guidelines for the OLAP system.

Before we get into a more detailed discussion of the major features of OLAP systems, let us list the most fundamental characteristics in plain language. OLAP systems

- Let business users have a multidimensional and logical view of the data in the data warehouse.
- Facilitate interactive query and complex analysis for the users.
- Allow users to drill down for greater details or roll up for aggregations of metrics along a single business dimension or across multiple dimensions.
- Provide the ability to perform intricate calculations and comparisons.
- Present results in a number of meaningful ways, including charts and graphs.

MAJOR FEATURES AND FUNCTIONS

Very often, you are faced with the question of whether OLAP is not just data warehousing in a nice wrapper? Can you not consider online analytical processing as just an information delivery technique and nothing more? Is it not another layer in the data warehouse, providing an interface between the data and the users? In some sense, OLAP is an information delivery system for the data warehouse. But OLAP is much more than that. A data warehouse

stores data and provides simpler access to business intelligence. An OLAP system complements the data warehouse by lifting the information delivery capabilities to new heights.

General Features

In this section, we will pay special attention to a few major features and functions of OLAP systems. You will gain greater insight into dimensional analysis, find deeper meanings about the necessity for drill downs and roll ups during analysis sessions and gain greater appreciation for the role of slicing and dicing operations in analysis. Before getting into greater details about these, let us recapitulate the general features of OLAP. Figure 15-4 summarizes these features. Also note the distinction between basic features and advanced features. The list shown in the figure includes the general features you observe in practice in most OLAP environments. Use the list as a quick checklist of features your project team must consider for your OLAP system.

Dimensional Analysis

By this time, you are perhaps tired of the term “dimensional analysis.” We have used the term many times so far. You have been told that dimensional analysis is a strong suit in the arsenal of OLAP. Any OLAP system devoid of multidimensional analysis is utterly useless. So try to get a clear picture of the facility provided in OLAP systems for dimensional analysis.

Let us begin with a simple STAR schema. This STAR schema has three business dimensions, namely, product, time, and store. The fact table contains sales. Figure 15-5 shows the STAR schema and a three-dimensional representation of the model as a cube, with products on the X-axis, time on the Y-axis, and stores on the Z-axis. What are the values represented along each axis? For example, in the STAR schema, time is one of the dimensions and month is one of the attributes of the time dimension. Values of this attribute month are

BASIC FEATURES	Multidimensional analysis	Consistent performance	Fast response times for interactive queries
	Drill-down and roll-up	Navigation in and out of details	Slice-and-dice or rotation
	Multiple view modes	Easy scalability	Time intelligence (year-to-date, fiscal period)
ADVANCED FEATURES	Powerful calculations	Cross-dimensional calculations	Pre-calculation or pre-consolidation
	Drill-through across dimensions or details	Sophisticated presentation & displays	Collaborative decision making
	Derived data values through formulas	Application of alert technology	Report generation with agent technology

Figure 15-4 General features of OLAP.

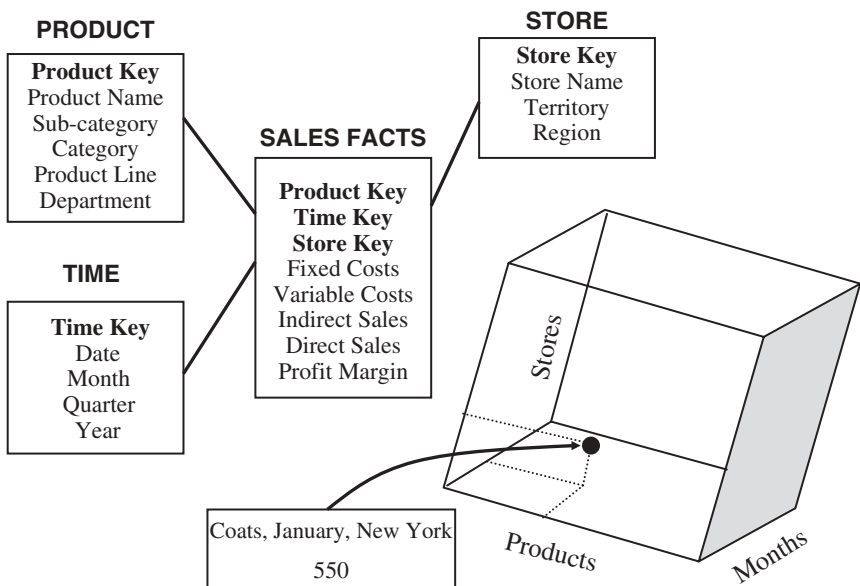


Figure 15-5 Simple STAR schema.

represented on the Y-axis. Similarly, values of the attributes product name and store name are represented on the other two axes.

This schema with just three business dimensions does not even look like a star. Nevertheless, it is a dimensional model. From the attributes of the dimension tables, pick the attribute product name from the product dimension, month from the time dimension, and store name from the store dimension. Now look at the cube representing the values of these attributes along the primary edges of the physical cube. Go further and visualize the sales for coats in the month of January at the New York store to be at the intersection of the three lines representing the product: coats, month: January, and store: New York.

If you are displaying the data for sales along these three dimensions on a spreadsheet, the columns may display the product names, the rows the months, and pages the data along the third dimension of store names. See Figure 15-6, which shows a screen display of a page of this three-dimensional data.

The page displayed on the screen shows a slice of the cube. Now look at the cube and move along a slice or plane passing through the point on the Z-axis representing store: New York. The intersection points on this slice or plane relate to sales along product and time business dimensions for store: New York. Try to relate these sale numbers to the slice on the cube representing store: New York.

Now we have a way of depicting three business dimensions and a single fact on a two-dimensional page and also on a three-dimensional cube. The numbers in each cell on the page are the sale numbers. What could be the types of multidimensional analysis on this particular set of data? What types of queries could be run during the course of analysis sessions? You could get sale numbers along the hierarchies of a combination of the three business dimensions of product, store, and time. You could perform various types of three-dimensional analysis of sales. The results of queries during analysis sessions will be displayed on the screen with the three dimensions represented in columns, rows, and

Store: New York

Products

PAGES: STORE dimensionCOLUMNS: PRODUCT dimension

ROWS: TIME dimension Months		Hats	Coats	Jackets	Dresses	Shirts	Slacks
	Jan	200	550	350	500	520	490
	Feb	210	480	390	510	530	500
	Mar	190	480	380	480	500	470
	Apr	190	430	350	490	510	480
	May	160	530	320	530	550	520
	Jun	150	450	310	540	560	330
	Jul	130	480	270	550	570	250
	Aug	140	570	250	650	670	230
	Sep	160	470	240	630	650	210
	Oct	170	480	260	610	630	250
	Nov	180	520	280	680	700	260
	Dec	200	560	320	750	770	310

Figure 15-6 A three-dimensional display.

pages. The following is a sample of simple queries and the result sets during a multidimensional analysis session.

Query

Display the total sales of all products for past five years in all stores.

Display of Results

Rows: Year numbers 2009, 2008, 2007, 2006, 2005

Columns: Total Sales for all products

Page: One store per page

Query

Compare total sales for all stores, product by product, between years 2009 and 2008.

Display of Results

Rows: Year numbers 2009, 2008; difference; percentage increase or decrease

Columns: One column per product, showing all products

Page: All stores

Query

Show comparison of total sales for all stores, product by product, between years 2009 and 2008 only for those products with reduced sales.

Display of Results

Rows: Year numbers 2009, 2008; difference; percentage decrease

Columns: One column per product, showing only the qualifying products

Page: All stores

Query

Show comparison of sales by individual stores, product by product, between years 2009 and 2008 only for those products with reduced sales.

Display of Results

Rows: Year numbers 2009, 2008; difference; percentage decrease

Columns: One column per product, showing only the qualifying products

Page: One store per page

Query

Show the results of the previous query, but rotating and switching the columns with rows.

Display of Results

Rows: One row per product, showing only the qualifying products

Columns: Year numbers 2009, 2008; difference; percentage decrease

Page: One store per page

Query

Show the results of the previous query, but rotating and switching the pages with rows.

Display of Results

Rows: One row per store

Columns: Year numbers 2009, 2008; difference; percentage decrease

Page: One product per page, displaying only the qualifying products.

This multidimensional analysis can continue on until the analyst determines how many products showed reduced sales and which stores suffered the most.

In the above example, we had only three business dimensions and each of the dimensions could, therefore, be represented along the edges of a cube or the results displayed as columns, rows, and pages. Now add another business dimension, promotion. That will bring the number of business dimensions to four. When you have three business dimensions, you are able to represent these three as a cube with each edge of the cube denoting one dimension. You are also able to display the data on a spreadsheet with two dimensions as rows and columns and the third dimension as pages. But when you have four dimensions or more, how can you represent the data? Obviously, a three-dimensional cube does not work. And you also have a problem when trying to display the data on a spreadsheet as rows, columns, and pages. So what about multidimensional analysis when there are more than three dimensions? This leads us to a discussion of hypercubes.

What Are Hypercubes?

Let us begin with the two business dimensions of product and time. Usually, business users wish to analyze not just sales but other metrics as well. Assume that the metrics to be

PRODUCT: Coats

PAGES: PRODUCT dimension COLUMNS: Metrics

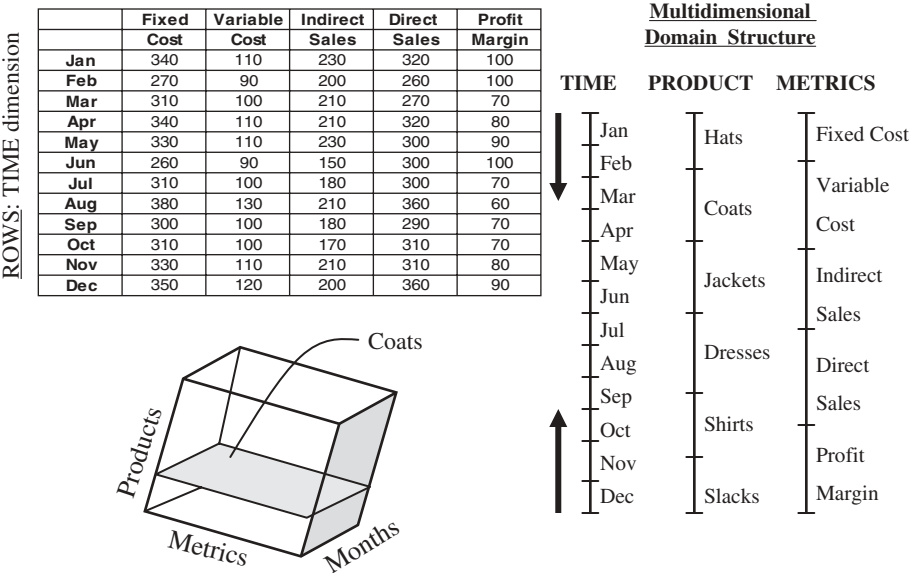


Figure 15-7 Display of columns, rows, and pages.

analyzed are fixed cost, variable cost, indirect sales, direct sales, and profit margin. These are five common metrics.

The data described here may be displayed on a spreadsheet showing metrics as columns, time as rows, and products as pages. Figure 15-7 shows a sample page of the spreadsheet display. In the figure, note the three straight lines, two of which represent the two business dimensions and the third, the metrics. You can independently move up or down along the straight lines. Some experts refer to this representation as a multidimensional domain structure (MDS).

The figure also shows a cube representing the data points along the edges. Relate the three straight lines to the three edges of the physical cube. Now the page you see in the figure is a slice passing through a single product and the divisions along the other two straight lines shown on the page as columns and rows. With three groups of data—two groups of business dimensions and one group of metrics—we can easily visualize the data as being along the three edges of a cube.

Now add another business dimension to the model. Let us add the store dimension. That results in three business dimensions plus the metrics data. How can you represent these four groups as edges of a three-dimensional cube? How do you represent a four-dimensional model with data points along the edges of a three-dimensional cube? How do you slice the data to display pages?

This is where an MDS diagram comes in handy. Now you need not try to perceive four-dimensional data as along the edges of the three-dimensional cube. All you have to do is draw four straight lines to represent the data as an MDS. These four lines represent the data (see Fig. 15-8). Looking at this figure, you realize that the metaphor of a physical cube to represent data breaks down when you try to represent four dimensions. But, as

Multidimensional
Domain Structure

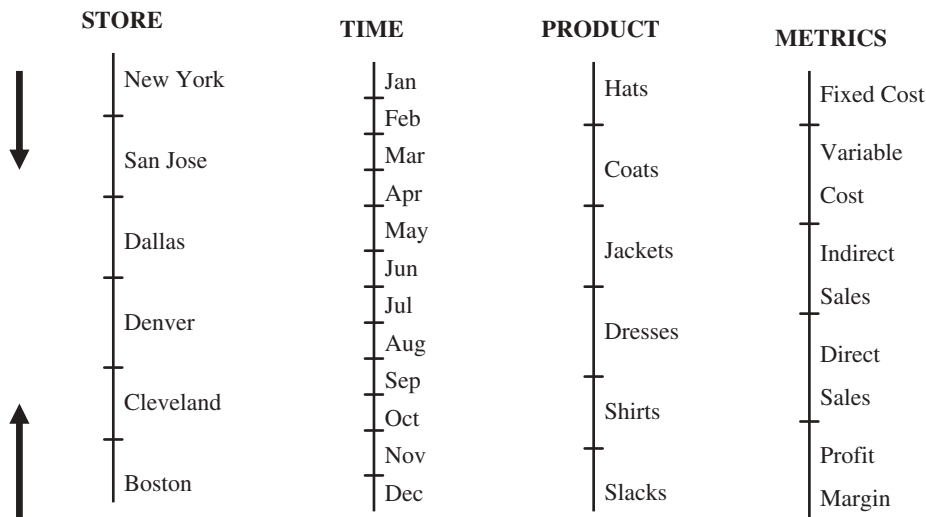


Figure 15-8 MDS for four dimensions.

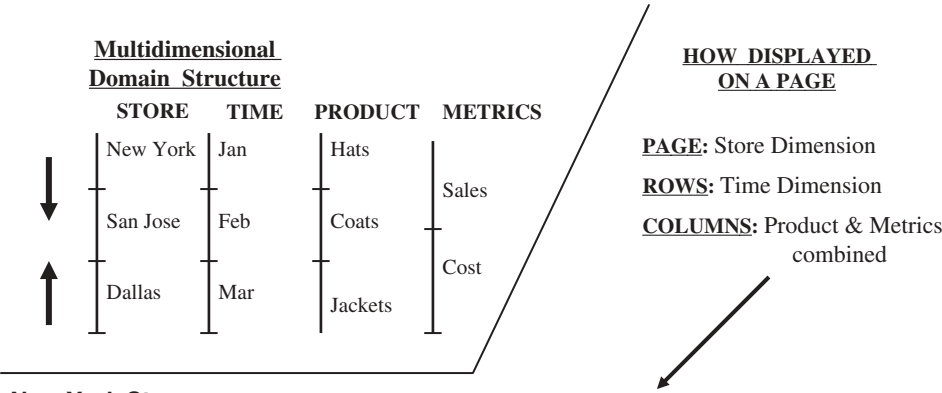
you see, the MDS is well suited to represent four dimensions. Can you think of the four straight lines of the MDS intuitively to represent a “cube” with four primary edges? This intuitive representation is a hypercube, a representation that accommodates more than three dimensions. At a lower level of simplification, a hypercube can very well accommodate three dimensions. A hypercube is a general metaphor for representing multidimensional data.

You now have a way of representing four dimensions as a hypercube. The next question relates to display of four-dimensional data on the screen. How can you possibly show four dimensions with only three display groups of rows, columns, and pages? Please turn your attention to Figure 15-9. What do you notice about the display groups? How does the display resolve the problem of accommodating four dimensions with only three display groups? By combining multiple logical dimensions within the same display group. Notice how product and metrics are combined to display as columns. The displayed page represents the sales for store: New York.

Let us look at just one more example of an MDS representing a hypercube. Let us move up to six dimensions. Please study Figure 15-10 with six straight lines showing the data representations. The dimensions shown in this figure are product, time, store, promotion, customer demographics, and metrics.

There are several ways you can display six-dimensional data on the screen. Figure 15-11 illustrates one such six-dimensional display. Notice how product and metrics are combined and represented as columns, store and time are combined as rows, and demographics and promotion as pages.

We have reviewed two specific issues. First, we have noted a special method for representing a data model with more than three dimensions using an MDS. This method is an intuitive way of showing a hypercube. A model with three dimensions can be represented



New York Store						
	Hats:Sales	Hats:Cost	Coats:Sales	Costs:Cost	Jackets:Sales	Jackets:Cost
Jan	450	350	550	450	500	400
Feb	380	280	460	360	400	320
Mar	400	310	480	410	450	400

Figure 15-9 Page displays for four-dimensional data.

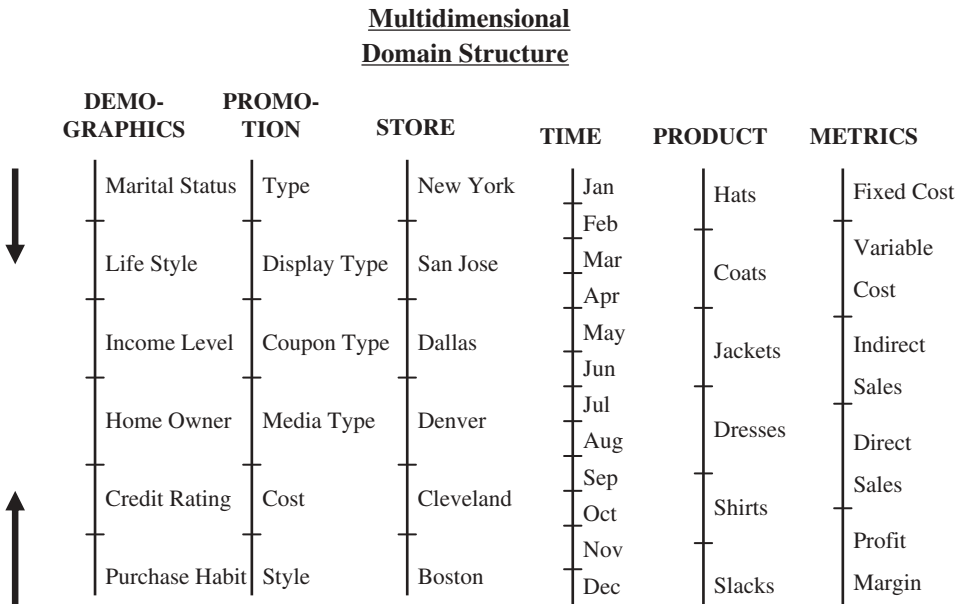


Figure 15-10 Six-dimensional MDS.

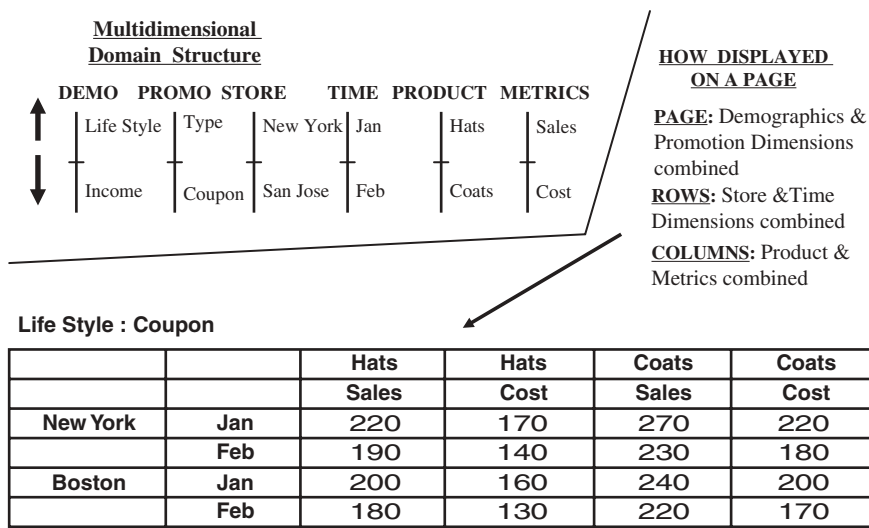


Figure 15-11 Page displays for six-dimensional data.

by a physical cube. But a physical cube is limited to only three dimensions or less. Second, we have also discussed the methods for displaying the data on a flat screen when the number of dimensions is three or more. Building on the resolution of these two issues, let us now move on to two very significant aspects of multidimensional analysis. One of these is the drill-down and roll-up exercise; the other is the slice-and-dice operation.

Drill Down and Roll Up

Return to Figure 15-5. Look at the attributes of the product dimension table of the STAR schema. In particular, note these specific attributes of the product dimension: product name, subcategory, category, product line, and department. These attributes signify an ascending hierarchical sequence from product name to department. A department includes product lines, a product line includes categories, a category includes subcategories, and each subcategory consists of products with individual product names. In an OLAP system, these attributes are called hierarchies of the product dimension.

OLAP systems provide drill-down and roll-up capabilities. Try to understand what we mean by these capabilities with reference to the above example. Figure 15-12 illustrates these capabilities with reference to the product dimension hierarchies. Note the different types of information given in the figure. It shows the rolling up to higher hierarchical levels of aggregation and the drilling down to lower levels of detail. Also note the sales numbers shown alongside. These are sales for one particular store in one particular month at these levels of aggregation. The sale numbers you notice as you go down the hierarchy are for a single department, a single product line, a single category, and so on. You drill down to get the lower level breakdown of sales. The figure also shows the drill across to another OLAP summarization using a different set of hierarchies of other dimensions. Notice also the drill through to the lower levels of granularity, as stored in the source data warehouse repository. Roll up, drill down, drill across, and drill through are extremely useful features of OLAP systems supporting multidimensional analysis.

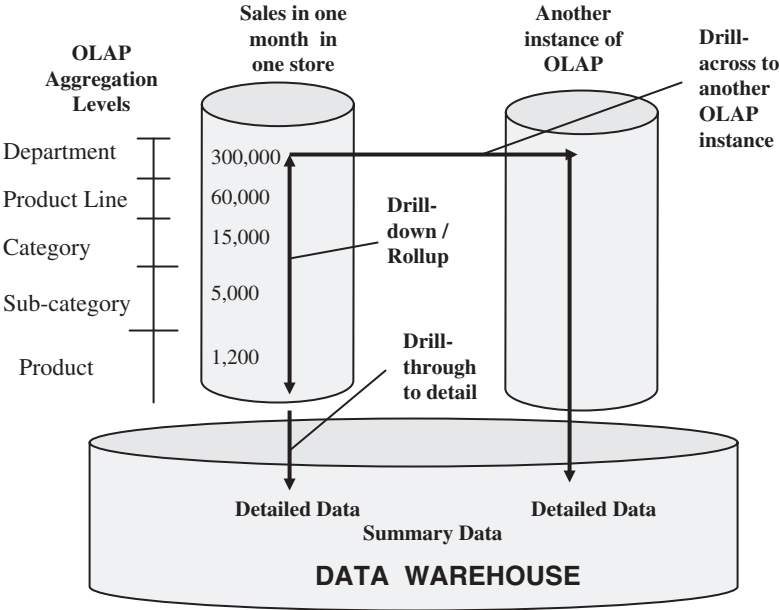


Figure 15-12 Roll-up and drill-down features of OLAP.

One more question remains. While you are rolling up or drilling down, how do the page displays change on the spreadsheets? For example, return to Figure 15-6 and look at the page display on the spreadsheet. The columns represent the various products, the rows represent the months, and the pages represent the stores. At this point, if you want to roll up to the next higher level of subcategory, how will the display in Figure 15-6 change? The columns on the display will have to change to represent subcategories instead of products. Figure 15-13 indicates this change.

Store: New York

Sub-categories

PAGES: STORE dimension

COLUMNS: PRODUCT dimension

ROWS: TIME dimension		Outer	Dress	Casual
	Jan	1,100	1,020	490
	Feb	1,080	1,040	500
	Mar	1,050	980	470
	Apr	970	1,000	480
	May	1,010	1,080	520
	Jun	910	1,100	330
	Jul	880	1,120	250
	Aug	960	1,320	230
	Sep	870	1,280	210
	Oct	910	1,240	250
	Nov	980	1,380	260
	Dec	1,080	1,520	310

Months

Figure 15-13 Three-dimensional display with roll-up.

Let us ask just one more question before we leave this subsection. When you have rolled up to the subcategory level in the product dimension, what happens to the display if you also roll up to the next higher level of the store dimension, territory? How will the display on the spreadsheet change? Now the spreadsheet will display the sales with columns representing subcategories, rows representing months, and the pages representing territories.

Slice and Dice or Rotation

Let us revisit Figure 15-6 showing the display of months as rows, products as columns, and stores as pages. Each page represents the sales for one store. The data model corresponds to a physical cube with these data elements represented by its primary edges. The page displayed is a slice or two-dimensional plane of the cube. In particular, this display page for the New York store is the slice parallel to the product and time axes. Now look at Figure 15-14 carefully. On the left side, the first part of the diagram shows this alignment of the cube. For the sake of simplicity, only three products, three months, and three stores are chosen for illustration.

Now rotate the cube so that products are along the Z-axis, months are along the X-axis, and stores are along the Y-axis. The slice we are considering also rotates. What happens to the display page that represents the slice? Months are now shown as columns and stores as rows. The display page represents the sales of one product, namely product: hats.

You can go to the next rotation so that months are along the Z-axis, stores are along the X-axis, and products are along the Y-axis. The slice we are considering also rotates. What happens to the display page that represents the slice? Stores are now shown as columns and products as rows. The display page represents the sales of one month, namely month: January.

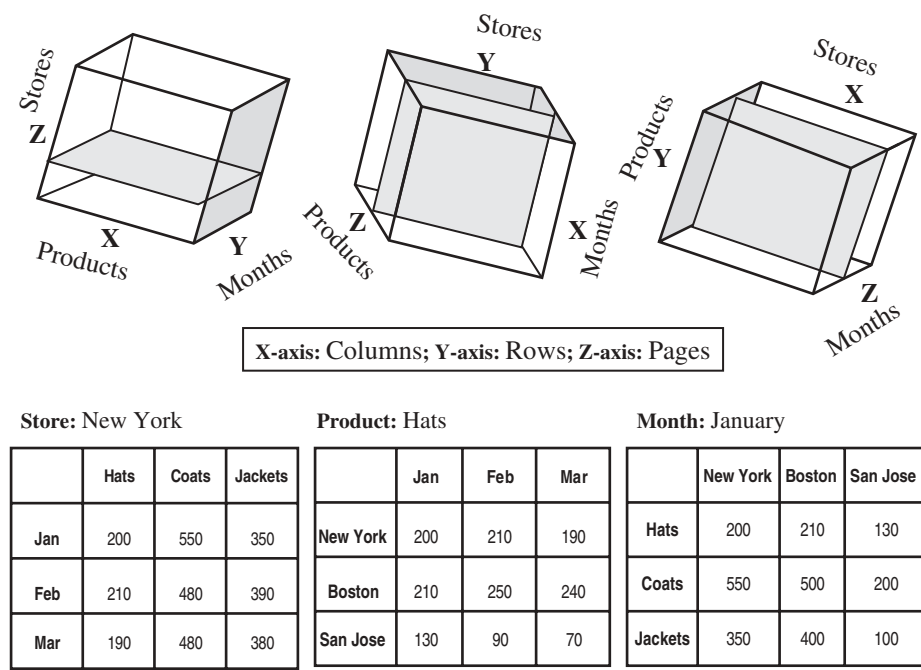


Figure 15-14 Slicing and dicing.

What is the great advantage of all of this for the users? Did you notice that with each rotation, the users can look at page displays representing different versions of the slices in the cube. The users can view the data from many angles, understand the numbers better, and arrive at meaningful conclusions.

Uses and Benefits

After exploring the features of OLAP in sufficient detail, you must have already deduced the enormous benefits of OLAP. We have discussed multidimensional analysis as provided in OLAP systems. The ability to perform multidimensional analysis with complex queries sometimes also entails complex calculations.

Let us summarize the benefits of OLAP systems:

- Increased productivity of business managers, executives, and analysts.
- Inherent flexibility of OLAP systems means that users may be self-sufficient in running their own analysis without IT assistance.
- Benefit for IT developers because using software specifically designed for the system development results in faster delivery of applications.
- Self-sufficiency for users, resulting in reduction in backlog.
- Faster delivery of applications following from the previous benefits.
- More efficient operations through reducing time on query executions and in network traffic.
- Ability to model real-world challenges with business metrics and dimensions.

OLAP MODELS

Have you heard of the terms ROLAP or MOLAP? How about the variation, DOLAP? A very simple explanation of the variations relates to the way data is stored for OLAP. The processing is still online analytical processing; basically, the storage methodology is different.

ROLAP. Refers to relational online analytical processing. In this case, the OLAP system is built on top of a relational database.

MOLAP. Refers to multidimensional online analytical processing. In this case, the OLAP system is implemented through a specialized multidimensional database.

HOLAP. Refers to hybrid online analytical processing. This models attempts to combine the strengths and features of ROLAP and MOLAP.

DOLAP. Refers for desktop online analytical processing. DOLAP is meant to provide portability to users of online analytical processing. In the DOLAP methodology, multidimensional datasets are created and transferred to the desktop machine, requiring only the DOLAP software to exist on that machine. DOLAP is a variation of ROLAP.

Database OLAP. Refers to a relational database management system (RDBMS) designated to support OLAP structures and to perform OLAP calculations.

Web OLAP. Refers to online analytical processing where OLAP data is accessible from a Web browser.

Overview of Variations

MOLAP and ROLAP are the fundamental models; therefore, we will discuss these in sufficient depth. In the MOLAP model, online analytical processing is best implemented by storing the data multidimensionally, that is, easily viewed in a multidimensional way. Here the data structure is fixed so that the logic to process multidimensional analysis can be based on well-defined methods of establishing data storage coordinates. Usually, multidimensional databases (MDDBs) are vendors' proprietary systems. On the other hand, the ROLAP model relies on the existing relational DBMS of the data warehouse. OLAP features are provided against the relational database.

See Figure 15-15 contrasting the two models. Notice the MOLAP model shown on the left side of the figure. The OLAP engine resides on a special server. Proprietary multidimensional databases (MDDBs) store data in the form of multidimensional hypercubes. You have to run special extraction and aggregation jobs from the relational database of the data warehouse to create these multidimensional data cubes in the MDDBs. The special server presents the data as OLAP cubes for processing by the users.

On the right side of the figure you see the ROLAP model. The OLAP engine resides on the desktop. Prefabricated multidimensional cubes are not created beforehand and stored in special databases. The relational data is presented as virtual multidimensional data cubes.

The MOLAP Model

As discussed, in the MOLAP model, data for analysis is stored in specialized multidimensional databases. Large multidimensional arrays form the storage structures. For example,

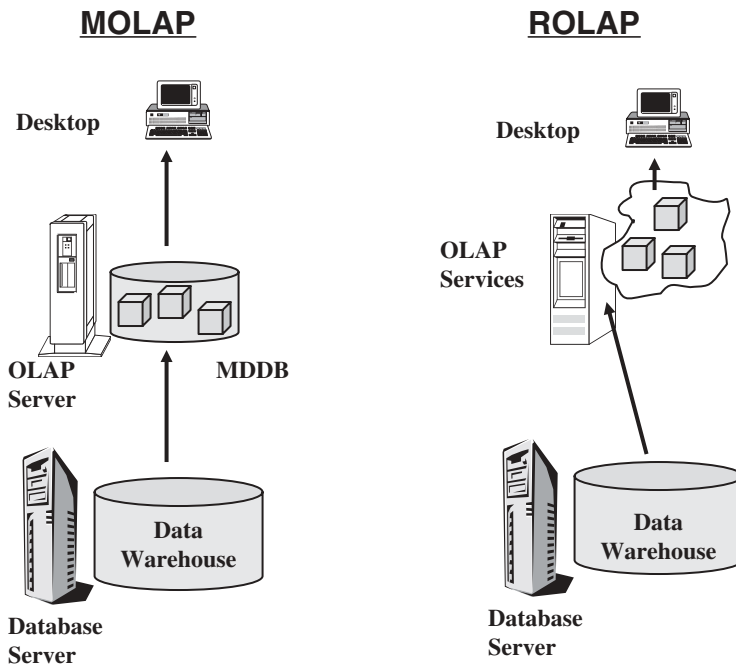


Figure 15-15 OLAP models.

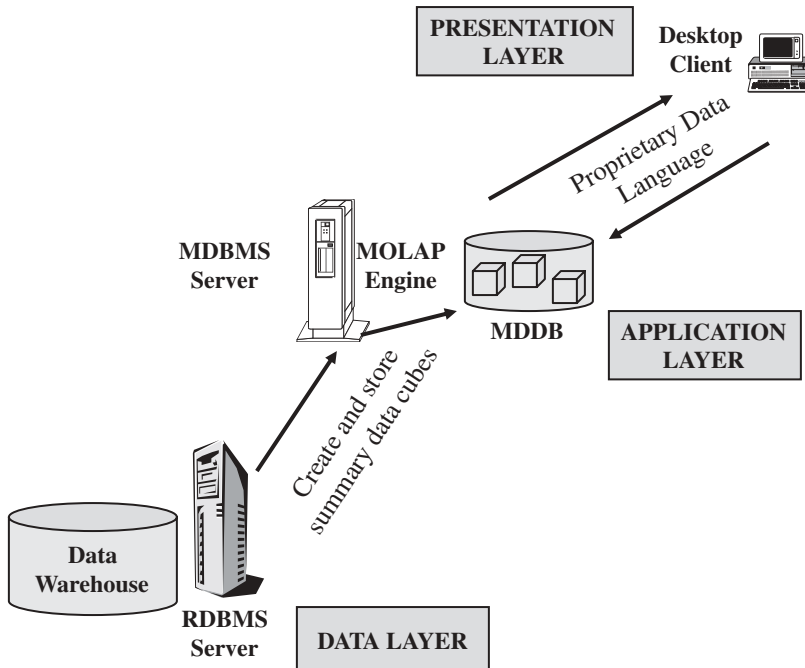


Figure 15-16 The MOLAP model.

to store sales number of 500 units for product ProductA, in month number 2009/01, in store StoreS1, under distributing channel Channe105, the sales number of 500 is stored in an array represented by the values (ProductA, 2009/01, StoreS1, Channe105).

The array values indicate the location of the cells. These cells are intersections of the values of dimension attributes. If you note how the cells are formed, you will realize that not all cells have values of metrics. If a store is closed on Sundays, then the cells representing Sundays will all be nulls.

Let us now consider the architecture for the MOLAP model. Please go over each part of Figure 15-16 carefully. Note the three layers in the multitier architecture. Precalculated and prefabricated multidimensional data cubes are stored in multidimensional databases. The MOLAP engine in the application layer pushes a multidimensional view of the data from the MDDBs to the users.

As mentioned earlier, multidimensional database management systems are proprietary software systems. These systems provide the capability to consolidate and fabricate summarized cubes during the process that loads data into the MDDBs from the main data warehouse. The users who need summarized data enjoy fast response times from the pre-consolidated data.

The ROLAP Model

In the ROLAP model, data is stored as rows and columns as in a relational data model. This model presents data to the users in the form of business dimensions. In order to hide the storage structure to the user and present data multidimensionally, a semantic layer of meta-data is created. The metadata layer supports the mapping of dimensions to the relational

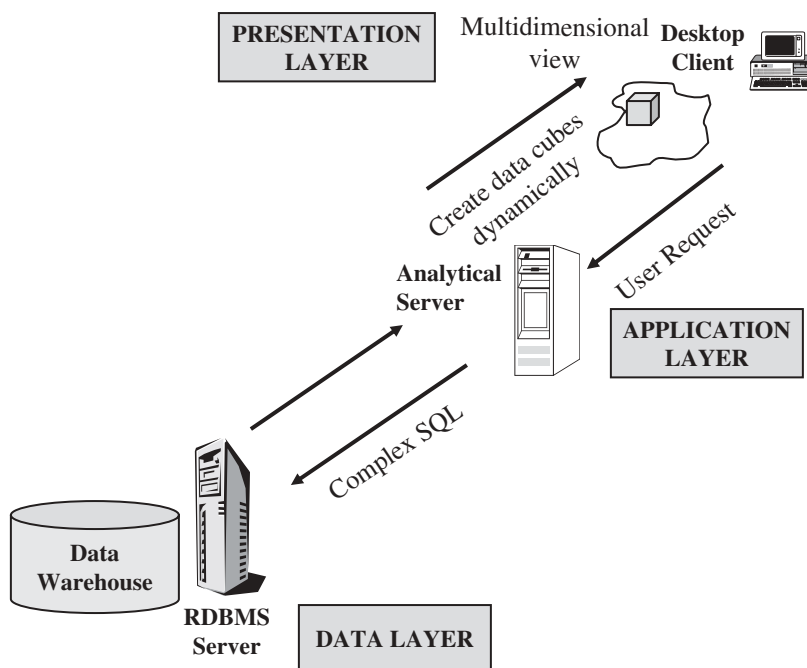


Figure 15-17 The ROLAP model.

tables. Additional metadata supports summarizations and aggregations. You may store the meta-data in relational databases.

Now see Figure 15-17. This figure shows the architecture of the ROLAP model. What you see is a three-tier architecture. The analytical server in the middle tier application layer creates multidimensional views on the fly. The multidimensional system at the presentation layer provides a multidimensional view of the data to the users. When the users issue complex queries based on this multidimensional view, the queries are transformed into complex SQL directed to the relational database. Unlike the MOLAP model, static multidimensional structures are not created and stored.

True ROLAP has three distinct characteristics:

- Supports all the basic OLAP features and functions discussed earlier
- Stores data in a relational form
- Supports some form of aggregation

Local hypercubing is a variation of ROLAP provided by vendors. This is how it works:

- 1 The user issues a query.
- 2 The results of the query get stored in a small, local, multidimensional database.
- 3 The user performs analysis against this local database.
- 4 If additional data is required to continue the analysis, the user issues another query and the analysis continues.

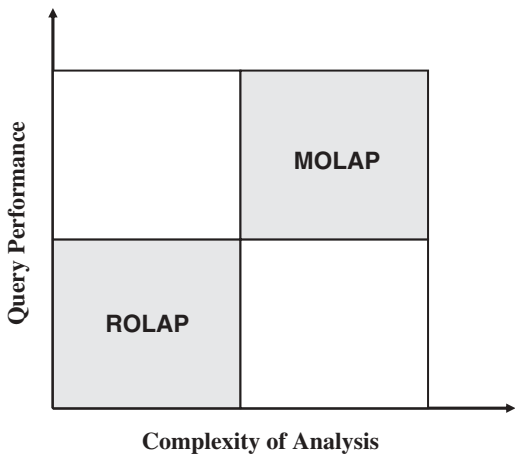


Figure 15-18 ROLAP or MOLAP?

ROLAP Versus MOLAP

Should you use the relational approach or the multidimensional approach to provide online analytical processing for your users? That depends on how important query performance is for your users. Again, the choice between ROLAP and MOLAP also depends on the complexity of the queries from your users. Figure 15-18 charts the solution options based

	Data Storage	Underlying Technologies	Functions and Features
ROLAP	Data stored as relational tables in the warehouse. Detailed and light summary data available. Very large data volumes. All data access from the warehouse storage.	Use of complex SQL to fetch data from warehouse. ROLAP engine in analytical server creates data cubes on the fly. Multidimensional views by presentation layer.	Known environment and availability of many tools. Limitations on complex analysis functions. Drill-through to lowest level easier. Drill-across not always easy.
MOLAP	Data stored as relational tables in the warehouse. Various summary data kept in proprietary databases (MDDBs) Moderate data volumes. Summary data access from MDDB, detailed data access from warehouse.	Creation of pre-fabricated data cubes by MOLAP engine. Propriety technology to store multidimensional views in arrays, not tables. High speed matrix data retrieval. Sparse matrix technology to manage data sparsity in summaries.	Faster access. Large library of functions for complex calculations. Easy analysis irrespective of the number of dimensions. Extensive drill-down and slice-and-dice capabilities.

Figure 15-19 ROLAP versus MOLAP.

on the considerations of query performance and complexity of queries. MOLAP is the choice for faster response and more intensive queries. These are just two broad considerations. Figure 15-19 provides a comprehensive comparison between the two.

OLAP IMPLEMENTATION CONSIDERATIONS

Before considering implementation of OLAP in your data warehouse, you have to take into account two key issues with regard to the MOLAP model running under MDDBMS. The first issue relates to the lack of standardization. Each vendor tool has its own client interface. Another issue is scalability. OLAP is generally good for handling summary data, but not good for volumes of detailed data.

On the other hand, highly normalized data in the data warehouse can give rise to processing overhead when you are performing complex analysis. You may reduce this by using a STAR schema multidimensional design. In fact, for some ROLAP tools, the multidimensional representation of data in a STAR schema arrangement is a prerequisite.

Consider a few choices of architecture. Look at Figure 15-20 showing four architectural options.

You have now studied the various implementation options for providing OLAP functionality in your data warehouse. These are important choices. Remember, without OLAP, your users have very limited means for analyzing data. Let us now examine some specific design considerations.

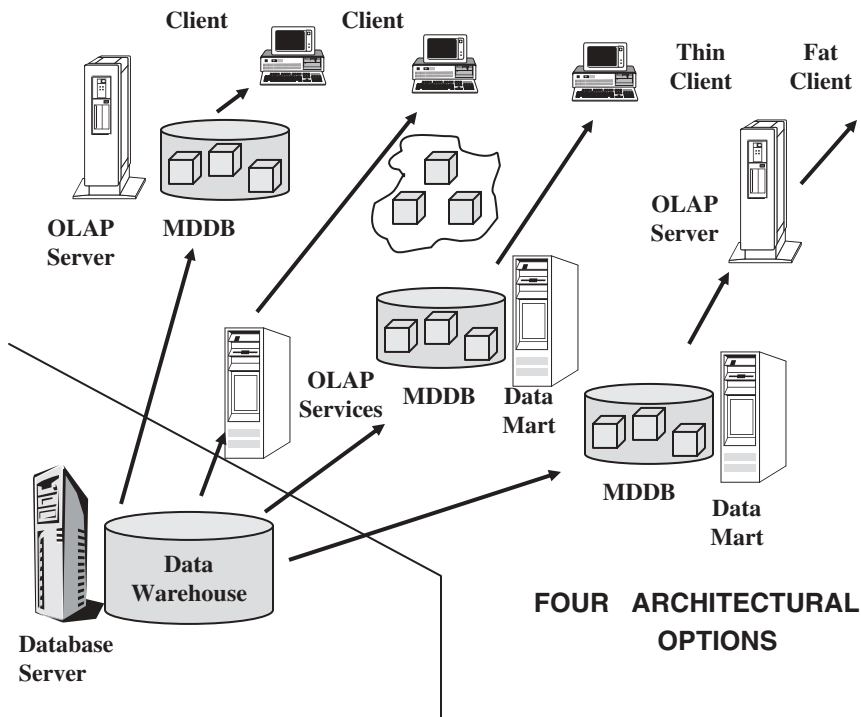


Figure 15-20 OLAP architectural options.

Data Design and Preparation

The data warehouse feeds data to the OLAP system. In the MOLAP model, separate proprietary multidimensional databases store the data fed from the data warehouse in the form of multidimensional cubes. On the other hand, in the ROLAP model, although no static intermediary data repository exists, data is still pushed into the OLAP system with cubes created dynamically on the fly. Thus, the sequence of the flow of data is from the operational source systems to the data warehouse and from there to the OLAP system.

Sometimes, you may have the desire to short-circuit the flow of data. You may wonder why you should not build the OLAP system on top of the operational source systems themselves. Why not extract data into the OLAP system directly? Why bother moving data into the data warehouse and then into the OLAP system? Here are a few reasons why this approach is flawed:

- An OLAP system needs transformed and integrated data. The system assumes that the data has been consolidated and cleansed somewhere before it arrives. The disparity among operational systems does not support data integration directly.
- The operational systems keep historical data only to a limited extent. An OLAP system needs extensive historical data. Historical data from the operational systems must be combined with archived historical data before it reaches the OLAP system.
- An OLAP system requires data in multidimensional representations. This calls for summarization in many different ways. Trying to extract and summarize data from the various operational systems at the same time is untenable. Data must be consolidated before it can be summarized at various levels and in different combinations.
- Assume there are a few OLAP systems in your environment. That is, one supports the marketing department, another supports the inventory control department, yet another supports the finance department, and so on. To accomplish this, you have to build a separate interface with the operational systems for data extraction into each OLAP system. Can you imagine how difficult this would be?

In order to help prepare the data for the OLAP system, let us first examine some significant characteristics of data in this system. Please review the following list:

- An OLAP system stores and uses much less data compared to a data warehouse.
- Data in the OLAP system is summarized. You will rarely find data at the lowest level of detail as in the data warehouse.
- OLAP data is more flexible for processing and analysis partly because there is much less data to work with.
- Every instance of the OLAP system in your environment is customized for the purpose that instance of the system serves.

In other words, OLAP data tends to be more departmentalized, whereas data in the data warehouse serves corporate-wide needs.

An overriding principle is that OLAP data is generally customized. When you build the OLAP system with system instances servicing different user groups, you need to keep this in mind. For example, one instance or specific set of summarizations would be meant for one group of users, say, the marketing department. Let us quickly go through the techniques for

preparing OLAP data for a specific group of users or a particular department, for example, marketing.

Define Subset Select the subset of detailed data the marketing department is interested in.

Summarize Summarize and prepare aggregate data structures in the way the marketing department needs for summarizing. For example, summarize products along product categories as defined by marketing. Sometimes, marketing and accounting departments may categorize products in different ways.

Denormalize Combine relational tables in exactly the same way the marketing department needs denormalized data. If marketing needs tables A and B joined, but finance needs tables B and C joined, go with the join for tables A and B for the marketing OLAP subset.

Calculate and Derive If some calculations and derivations of the metrics are department-specific in your company, use the ones for marketing.

Index Choose those attributes that are appropriate for marketing to build indexes.

What about data modeling for the OLAP data structure? The OLAP structure contains several levels of summarization and a few kinds of detailed data. How do you model these levels of summarization?

Please see Figure 15-21 indicating the types and levels of data in OLAP systems. These types and levels must be taken into consideration while performing data modeling for the OLAP systems. Pay attention to the different types of data in an OLAP system. When you model the data structures for your OLAP system, you need to provide for these types of data.

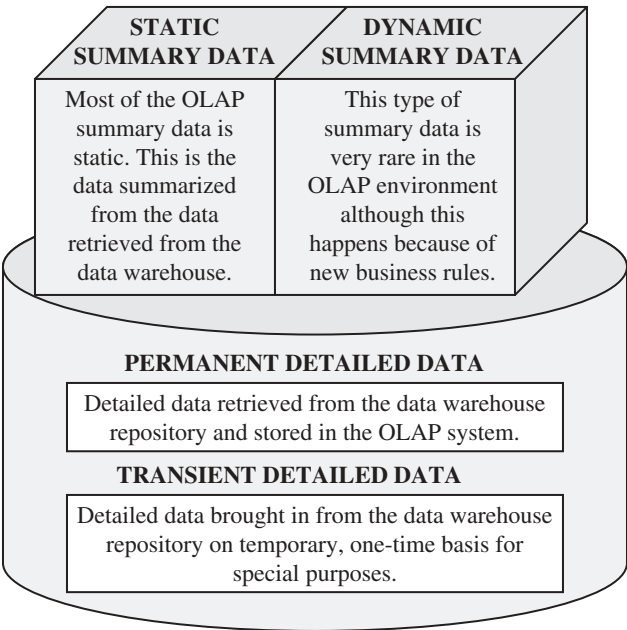


Figure 15-21 Data modeling considerations for OLAP.

Administration and Performance

Let us now turn our attention to two important though not directly connected issues.

Administration One of these issues is the matter of administration and management of the OLAP environment. The OLAP system is part of the overall data warehouse environment and, therefore, administration of the OLAP system is part of the data warehouse administration. Nevertheless, we must recognize some key considerations for administering and managing the OLAP system. Let us briefly indicate a few of these considerations.

- Expectations on what data will be accessed and how
- Selection of the right business dimensions
- Selection of the right filters for loading the data from the data warehouse
- Methods and techniques for moving data into the OLAP system (MOLAP model)
- Choosing the aggregation, summarization, and precalculation
- Developing application programs using the proprietary software of the OLAP vendor
- Size of the multidimensional database
- Handling of the sparse-matrix feature of multidimensional structures
- Drill down to the lowest level of detail
- Drill through to the data warehouse or to the source systems
- Drill across among OLAP system instances
- Access and security privileges
- Backup and restore facilities

Performance First you need to recognize that the presence of an OLAP system in your data warehouse environment shifts the workload. Some of the queries that usually must run against the data warehouse will now be redistributed to the OLAP system. The types of queries that need OLAP are complex and filled with involved calculations. Long and complicated analysis sessions consist of such complex queries. Therefore, when such queries get directed to the OLAP system, the workload on the main data warehouse becomes substantially reduced.

A corollary of shifting the complex queries to the OLAP system is the improvement in the overall query performance. The OLAP system is designed for complex queries. When such queries run in the OLAP system, they run faster. As the size of the data warehouse grows, the size of the OLAP system still remains manageable and comparably small.

Multidimensional databases provide a reasonably predictable, fast, and consistent response to every complex query. This is mainly because OLAP systems preaggregate and precalculate many, if not, all possible hypercubes and store these. The queries run against the most appropriate hypercubes. For instance, assume that there are only three dimensions. The OLAP system will calculate and store summaries as follows:

- A three-dimensional low-level array to store base data
- A two-dimensional array of data for dimension-1 and dimension-2
- A 2-dimensional array of data for dimension-2 and dimension-3
- A high-level summary array by dimension-1

- A high-level summary array by dimension-2
- A high-level summary array by dimension-3

All of these precalculations and preaggregations result in faster response to queries at any level of summarization. But this speed and performance do not come without any cost. You pay the price to some extent in the load performance. OLAP systems are not refreshed daily for the simple reason that load times for precalculating and loading all the possible hypercubes are exorbitant. Enterprises use longer intervals between refreshes of their OLAP systems. Most OLAP systems are refreshed once a month.

OLAP Platforms

Where does the OLAP system physically reside? Should it be on the same platform as the main data warehouse? Should it be planned to be on a separate platform from the beginning? What about growth of the data warehouse and the OLAP system? How do the growth patterns affect the decision? These are some of the questions you need to answer as you provide OLAP capability to your users.

Usually, the data warehouse and the OLAP system start out on the same platform. When both are small, it is cost-justifiable to keep both on the same platform. Within a year, it is usual to find rapid growth in the main data warehouse. The trend normally continues. As this growth happens, you may want to think of moving the OLAP system to another platform to ease the congestion. But how exactly would you know whether to separate the platforms and when to do so? Here are some guidelines:

- When the size and usage of the main data warehouse escalate and reach the point where the warehouse requires all the resources of the common platform, start acting on the separation.
- If too many departments need the OLAP system, then the OLAP requires additional platforms to run.
- Users expect the OLAP system to be stable and perform well. The data refreshes to the OLAP system are much less frequent. Although this is true for the OLAP system, daily application of incremental loads and full refreshes of certain tables are needed for the main data warehouse. If these daily transactions applicable to the data warehouse begin to disrupt the stability and performance of the OLAP system, then move the OLAP system to another platform.
- Obviously, in decentralized enterprises with OLAP users spread out geographically, one or more separate platforms for the OLAP system become necessary.
- If users of one instance of the OLAP system want to stay away from the users of another, then separation of platforms needs to be looked into.
- If the chosen OLAP tools need a configuration different from the platform of the main data warehouse, then the OLAP system requires a separate platform, configured correctly.

OLAP Tools and Products

The OLAP market is becoming sophisticated. Many OLAP products have appeared and most of the recent products are quite successful. Quality and flexibility of the products have improved remarkably.

Before we provide a checklist to be used for evaluation of OLAP products, let us list a few broad guidelines:

- Let your applications and the users drive the selection of the OLAP products. Do not be carried away by flashy technology.
- Remember, your OLAP system will grow both in size and in the number of active users. Determine the scalability of the products before you choose.
- Consider how easy it is to administer the OLAP product.
- Performance and flexibility are key ingredients in the success of your OLAP system.
- As technology advances, the differences in the merits between ROLAP and MOLAP appear to be somewhat blurred. Do not worry too much about these two methods. Concentrate on the matching of the vendor product with your users' analytical requirements. Flashy technology does not always deliver.

Now let us get to the selection criteria for choosing OLAP tools and products. While you evaluate the products, use the following checklist and rate each product against each item on the checklist:

- Multidimensional representation of data
- Aggregation, summarization, precalculation, and derivations
- Formulas and complex calculations in an extensive library
- Cross-dimensional calculations
- Time intelligence such as year-to-date, current and past fiscal periods, moving averages, and moving totals
- Pivoting, cross-tabs, drill-down, and roll-up along single or multiple dimensions
- Interface of OLAP with applications and software such as spreadsheets, proprietary client tools, third-party tools, and 4GL environments.

Implementation Steps

At this point, perhaps your project team has been given the mandate to build and implement an OLAP system. You know the features and functions. You know the significance. You are also aware of the important considerations. How do you go about implementing OLAP? Let us summarize the key steps. These are the steps or activities at a very high level. Each step consists of several tasks to accomplish the objectives of that step. You will have to come up with the tasks based on the requirements of your environment. Here are the major steps:

- Dimensional modeling
- Design and building of the MDDB
- Selection of the data to be moved into the OLAP system
- Data acquisition or extraction for the OLAP system
- Data loading into the OLAP server
- Computation of data aggregation and derived data
- Implementation of application on the desktop
- Provision of user training

COMPANIES AND OLAP IMPLEMENTATION NOTES

Time Warner: Supports users in three continents with a strategic market planning and analysis system.

The World Bank: Performs complex statistical analyses on a mass of worldwide econometric data.

Hewlett-Packard: Provides speedy operational reports using a desktop OLAP over the corporate intranet to numerous users.

Sun Microsystems: Supports business planning with a networked OLAP tool, entirely Web-based.

Dun & Bradstreet: Provides a vital link between customers' corporate data and underlying detailed information.

British Airways: Reduces processing costs through better analysis using OLAP databases tied to new general ledger.

Barclays Bank: Manages risk on loans for maximum profit.

British Petroleum: Performs worldwide planning with its second generation OLAP application.

IBM Finance: Provides finance portal for cost-effective analysis, reporting, and performance management.

Subaru of America: Improves customer service through more effective allocation of inventory to franchised dealers.

GlaxoSmithKline: Performs international financial reporting through OLAP databases.

Deluxe Corp.: World's No. 1 check printer, performs more accurate forecasting through planning/analysis applications.

Source: Olapreport.com and vendor Web sites (paraphrased)

Figure 15-22 Typical OLAP implementations.

Examples of Typical Implementations

As OLAP is so very important for analysis to many organizations, we would like to list some examples of OLAP implementations. See Figure 15-22.

CHAPTER SUMMARY

- OLAP is critical because its multidimensional analysis, fast access, and powerful calculations exceed that of other analysis methods.
- OLAP is defined on the basis of Codd's initial twelve guidelines.
- OLAP characteristics include multidimensional view of the data, interactive and complex analysis facility, ability to perform intricate calculations, and fast response time.
- Dimensional analysis is not confined to three dimensions that can be represented by a physical cube. Hypercubes provide a method for representing views with more dimensions.
- ROLAP and MOLAP are the two major OLAP models. The difference between them lies in the way the basic data is stored. Ascertain which model is more suitable for your environment.
- OLAP tools have matured. Some RDBMSs include support for OLAP.

REVIEW QUESTIONS

1. Briefly explain multidimensional analysis.
2. Name any four key capabilities of an OLAP system.
3. State any five of Dr. Codd's guidelines for an OLAP system, giving a brief description for each.
4. What are hypercubes? How do they apply in an OLAP system?
5. What is meant by slice-and-dice? Give an example.
6. What are the essential differences between the MOLAP and ROLAP models? Also list a few similarities.
7. What are multidimensional databases? How do these store data?
8. Describe any one of the four OLAP architectural options.
9. Discuss two reasons why feeding data into the OLAP system directly from the source operational systems is not recommended.
10. Name any four factors for consideration in OLAP administration.

EXERCISES

1. Indicate if true or false:
 - A. OLAP facilitates interactive queries and complex uses.
 - B. A hypercube can be represented by the physical cube.
 - C. Slice-and-dice is the same as the rotation of the columns and rows in presentation of data.
 - D. DOLAP stands for departmental OLAP.
 - E. ROLAP systems store data in multidimensional, proprietary databases.
 - F. The essential difference between ROLAP and MOLAP is in the way data is stored.
 - G. OLAP systems need transformed and integrated data.
 - H. Data in an OLAP system is rarely summarized.
 - I. Multidimensional domain structure (MDS) can represent only up to six dimensions.
 - J. OLAP systems do not handle moving averages.
2. As a senior analyst on the project team of a publishing company exploring the options for a data warehouse, make a case for OLAP. Describe the merits of OLAP and how it will be essential in your environment.
3. Pick any six of Dr. Codd's initial guidelines for OLAP. Give your reasons why the selected six are important for OLAP.
4. You are asked to form a small team to evaluate the MOLAP and ROLAP models and make your recommendations. This is part of the data warehouse project for a large

manufacturer of heavy chemicals. Describe the criteria your team will use to make the evaluation and selection.

5. Your company is the largest producer of chicken products, selling to supermarkets, fast-food chains, and restaurants, and also exporting to many countries. The analysts from many offices worldwide expect to use the OLAP system when implemented. Discuss how the project team must select the platform for implementing OLAP for the company. Explain your assumptions.